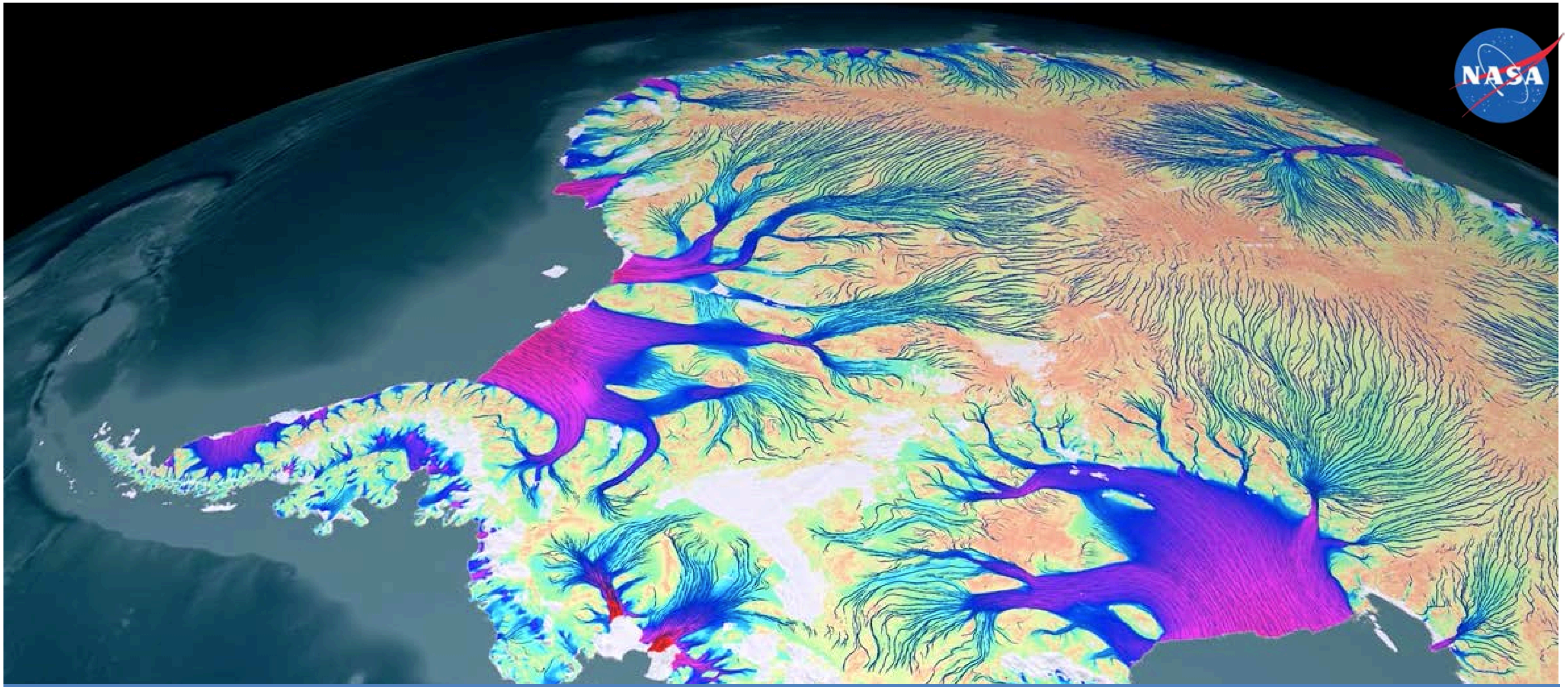
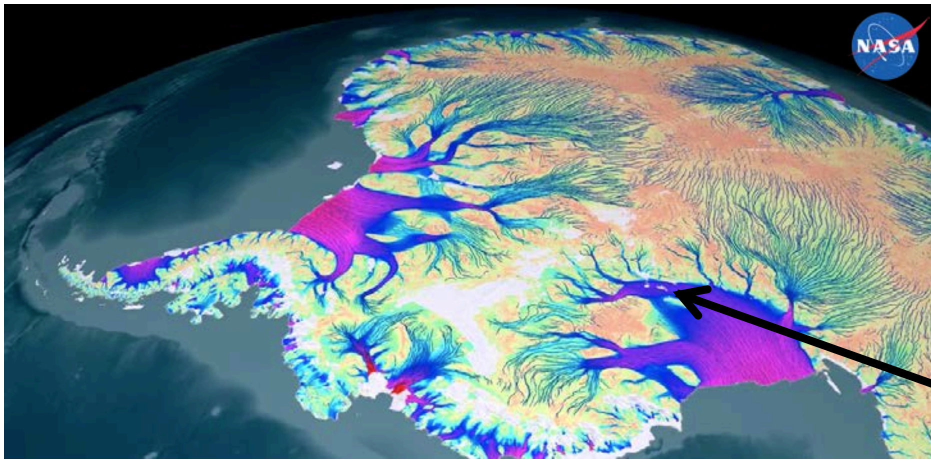


# The microseismicity of glacier sliding


F. Walter[1,2] and many others

**(1) Laboratory of Hydraulics, Hydrology and Glaciology (VAW), ETH Zürich, Switzerland**
**(2) Swiss Seismological Service, ETH Zürich, Switzerland**

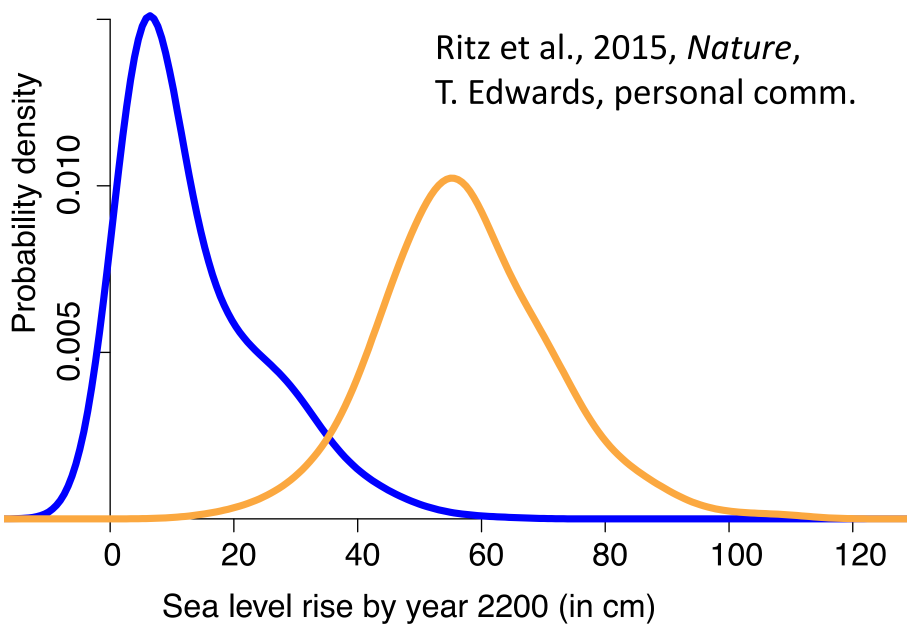
# Sliding Matters: Uncertainties in Sea Level Rise



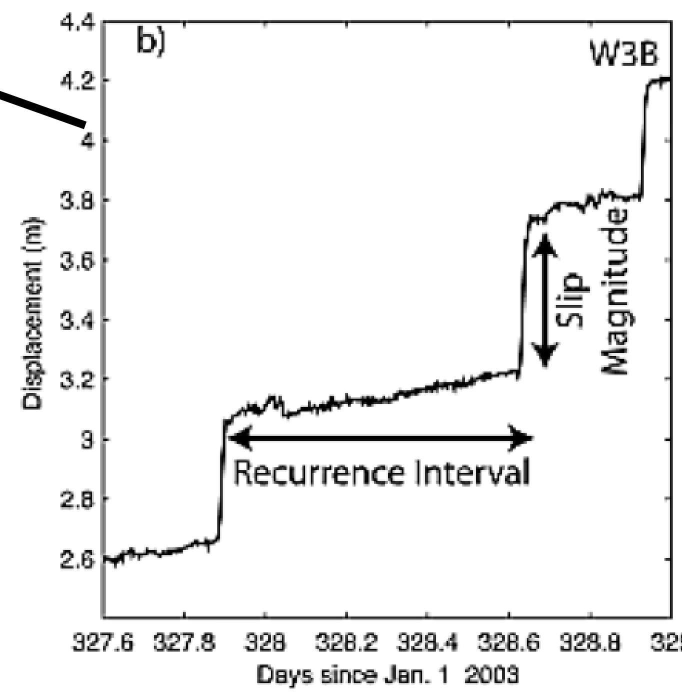

Linear viscous?

Plastic?

Or …?

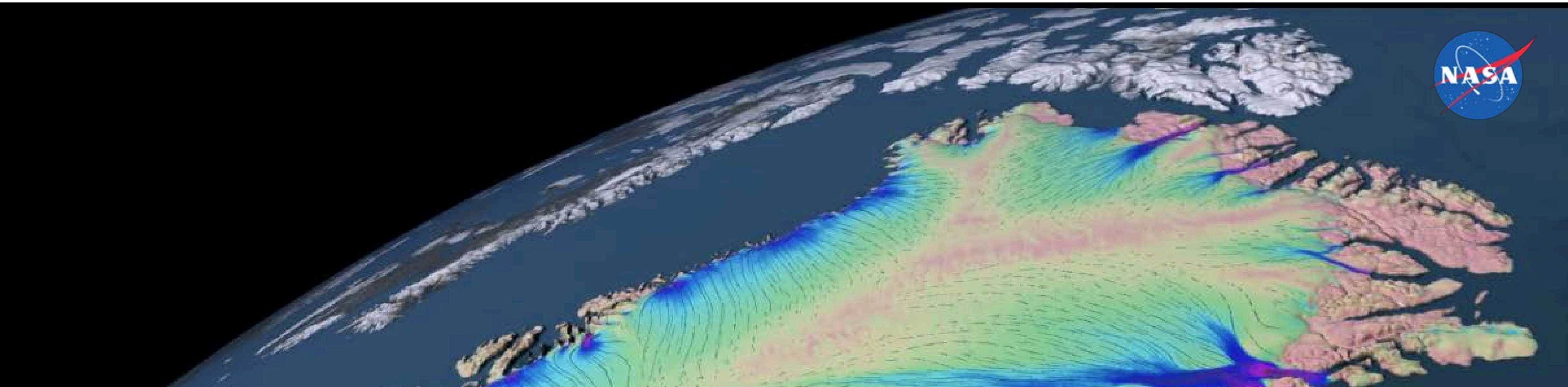


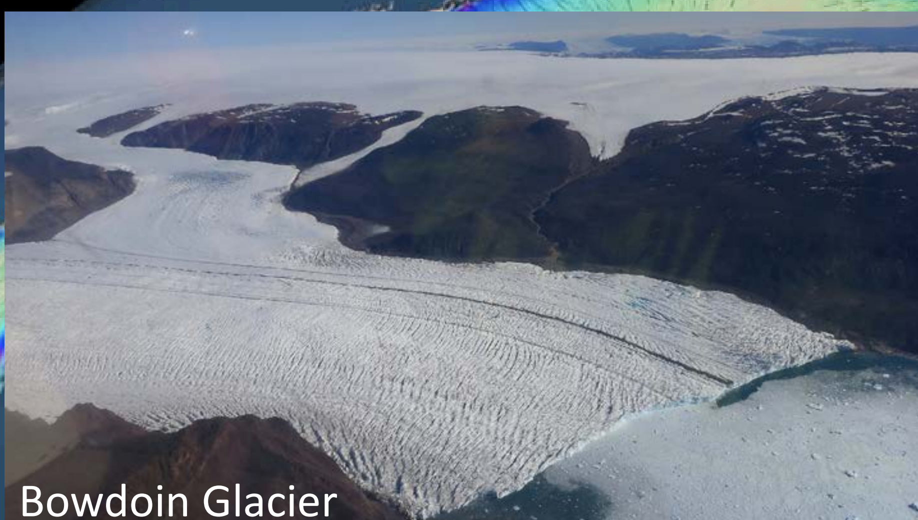

Ritz et al., 2015, *Nature*,
T. Edwards, personal comm.




Winberry et al., 2009, *JGR*

# Sliding Matters: “Everywhere”



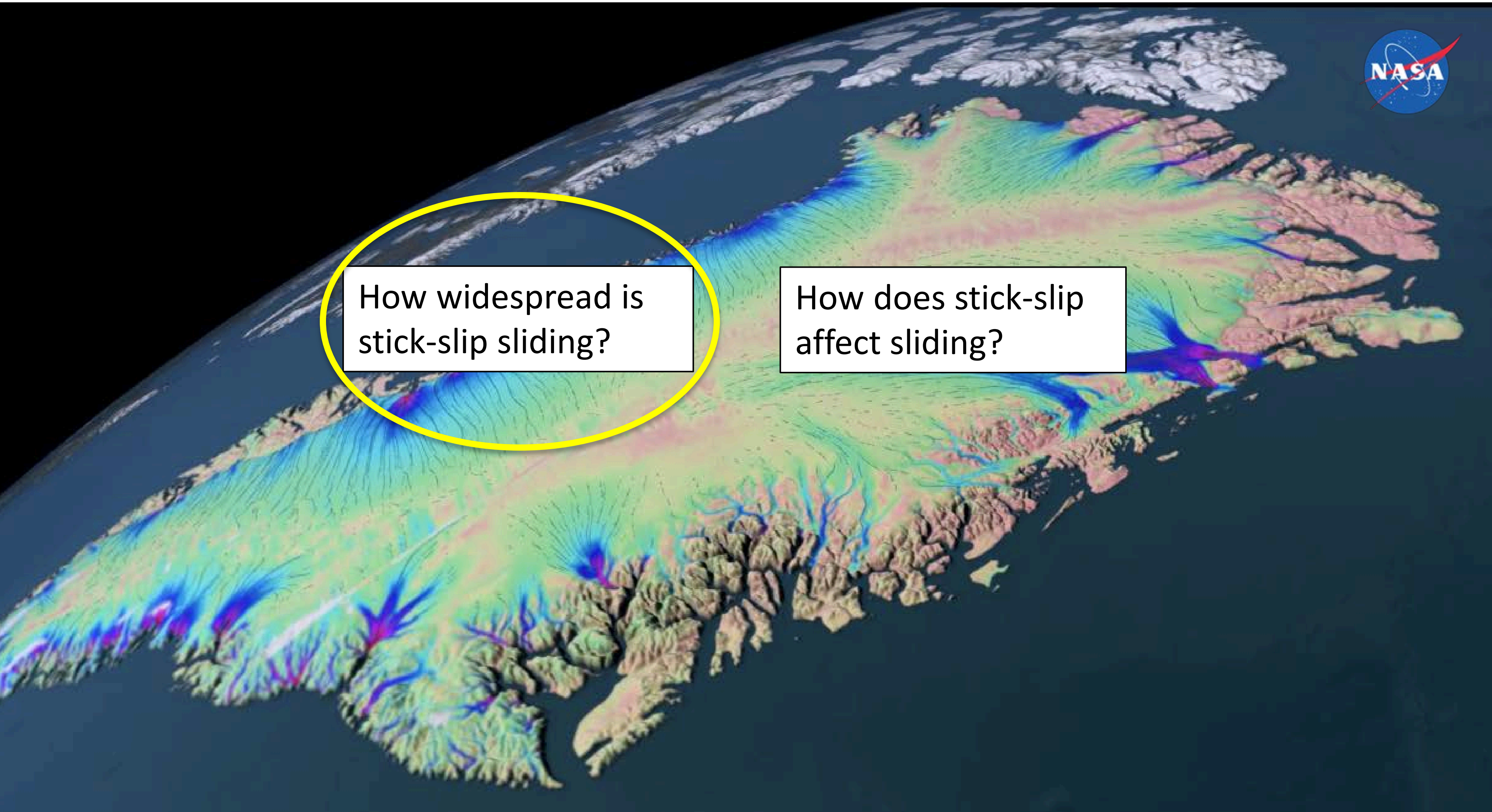



# Sliding Matters: “Everywhere”




See work by Lukas Preiswerk

# What about stick-slip?



# Stick-Slip in Antarctica

*Annals of Glaciology* 9 1987


**MICROEARTHQUAKES UNDER AND ALONGSIDE ICE STREAM B, ANTARCTICA, DETECTED BY A NEW PASSIVE SEISMIC ARRAY**

by

D.D. Blankenship, S. Anandakrishnan, J.L. Kempf and C.R. Bentley

(Geophysical and Polar Research Center, University of Wisconsin-Madison, Madison, WI 53706, U.S.A.)

- Since 1980’s
- Can make up all of ice motion
- Anatomy of slip episodes
- Ice fabric
- Erosion

# Stick-Slip in High-Melt (Alpine) Environments?

Journal of

GEOPHYS

VOLUME 75

Ic

Annals of Glaciology 54(64) 2013 doi:10.3189/2013AoG64A203 149

**The search for seismic signatures of movement at the glacier bed in a polythermal valley glacier**

AM[1]

h, UK

AGU PUBLICATIONS

JGR

**Journal of Geophysical Research: Earth Surface**

**RESEARCH ARTICLE**

10.1002/2014JF003288

This article is a companion to *Helmstetter et al.* [2015] doi:10.1002/2014JF003288.

**Basal icequakes recorded beneath an Alpine glacier (Glacier d'Argentière, Mont Blanc, France): Evidence for stick-slip motion?**

Agnès Helmstetter[1], Barbara Nicolas[2], Pierre Comon[2], and Michel Gay[2]

[1] University of Grenoble, CNRS, ISTerre, Grenoble, France, [2] Grenoble INP, GIPSA-Lab, CNRS, Grenoble, France

JOURN

**Deep icequakes: What happens at the base of Alpine glaciers?**

Fabian Walter,[1,2] Pierre Dalban Canassy,[2] Stephan Husen,[3] and John F. Clinton[3]

Received 10 April 2013; revised 23 July 2013; accepted 25 July 2013; published 5 September 2013.

andakrishnan[4], Miriam

USA

[2]Department of Geological Sciences, Central Washington University, Ellensburg, Washington 98926, USA

[3]Physics Department, St. Olaf College, Northfield, Minnesota 55057, USA

[4]Department of Geosciences, Penn State University, University Park, Pennsylvania 16802, USA

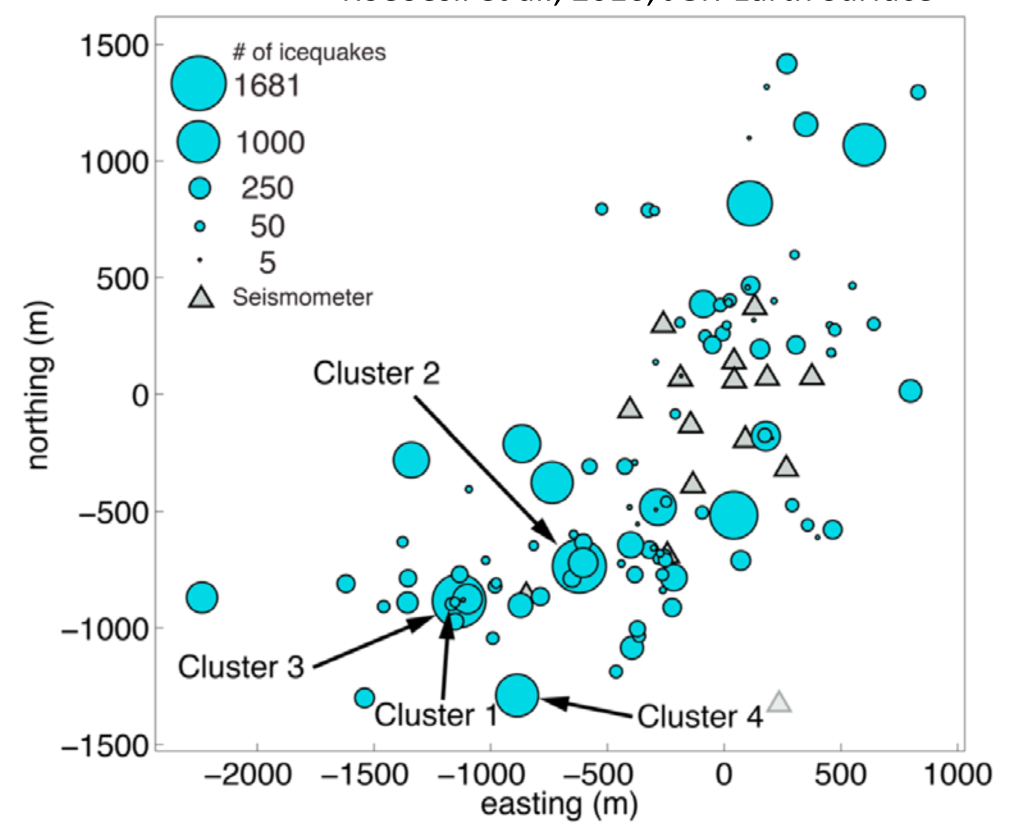
[5]Norwegian Water Resources and Energy Directorate, NO-0301 Oslo, Norway

Stick-Slip Icequakes

- Dominated by crevasse seismicity
- Weak signals
- Very high frequencies

Thickness: 400-600 m

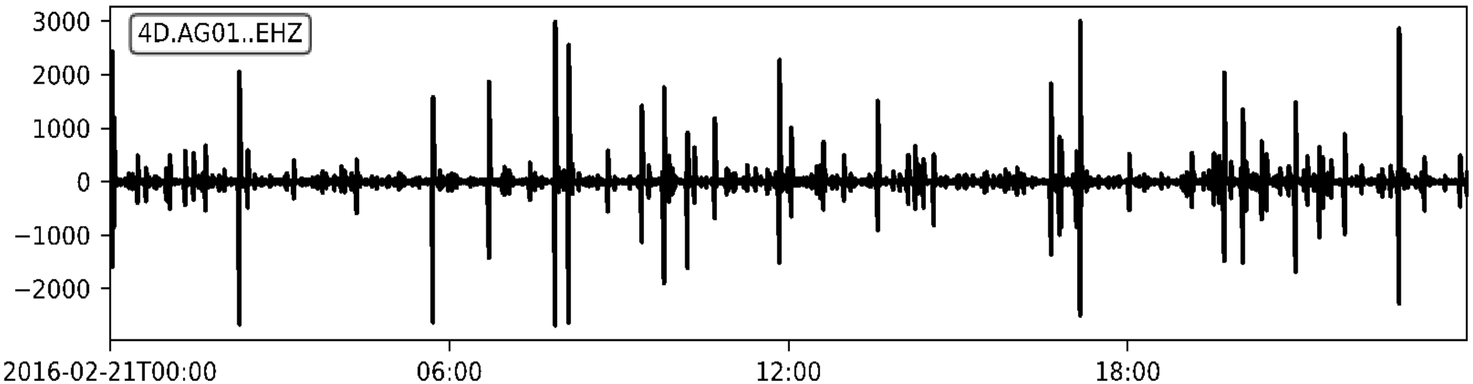
Roeoesli et al., 2016, JGR-Earth Surface



# Aletschgletscher

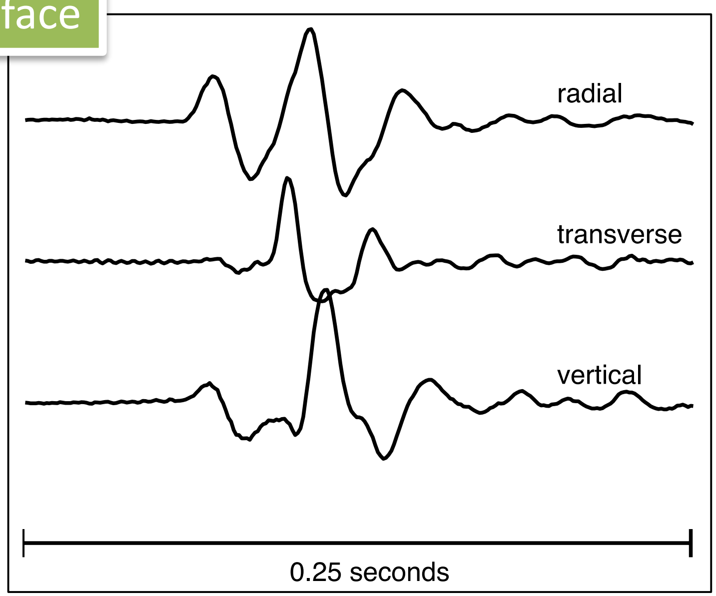
# Alpine Glacier Seismicity

High-frequency seismicity (>10 Hz) on winter day

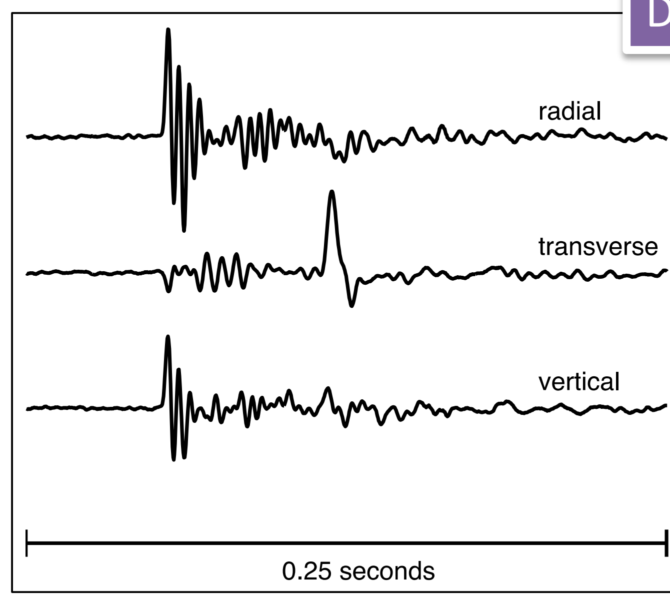


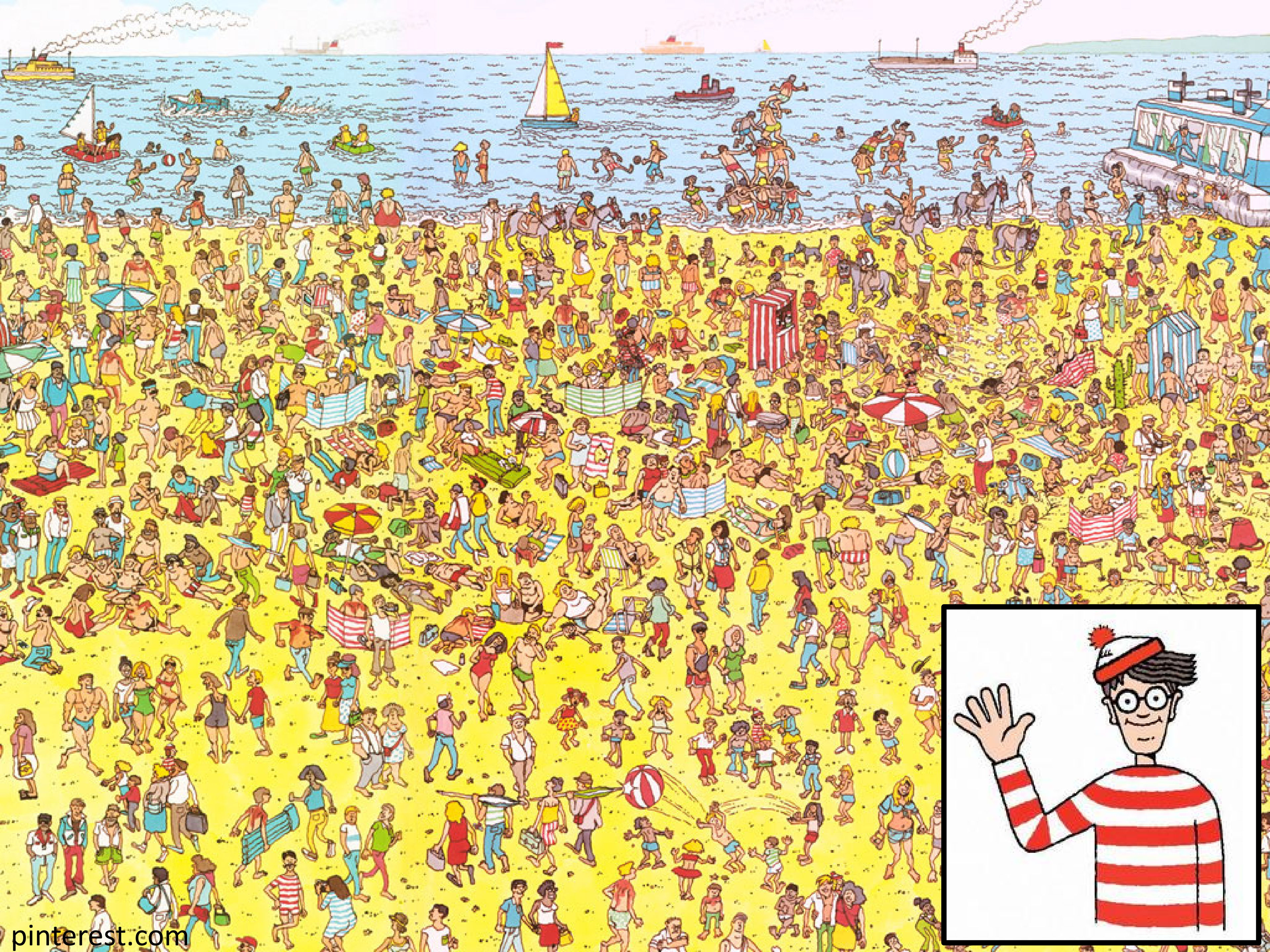

99 % surface seismicity

Surface




Deep



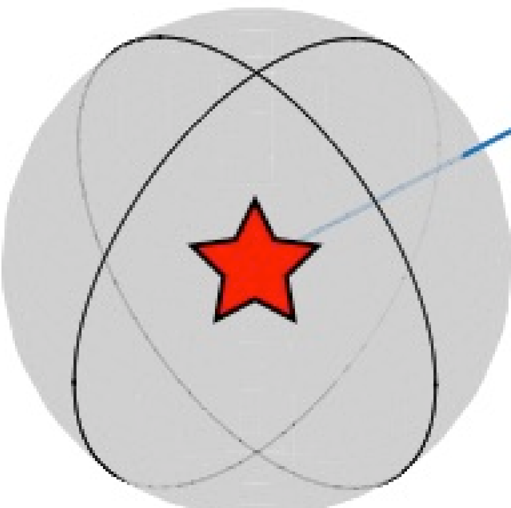


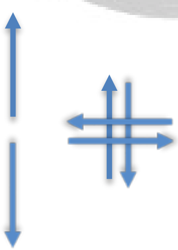
pinterest.com

# Shear Fault Seismicity

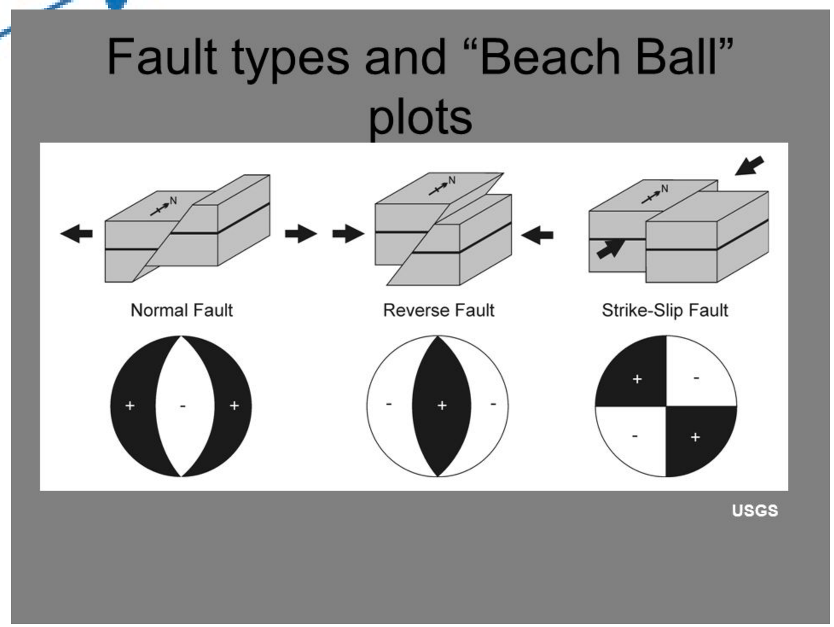
M= $\begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}$ General moment tensor

M= $\begin{pmatrix} M & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M \end{pmatrix}$ Shear fault

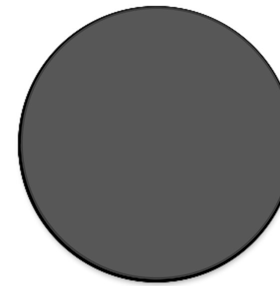




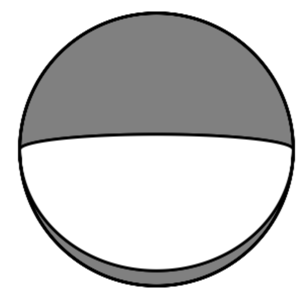
Force couple representation




Tensile Fault

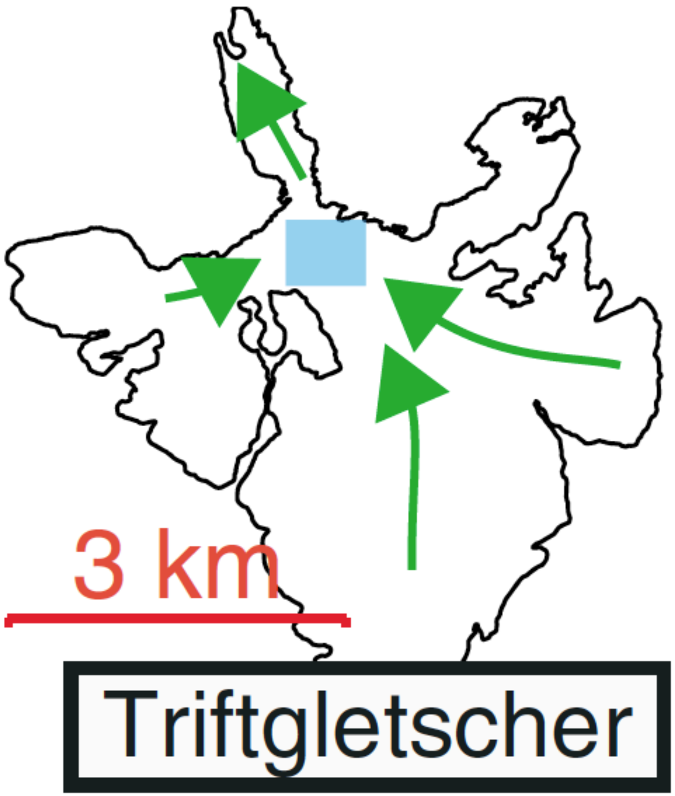


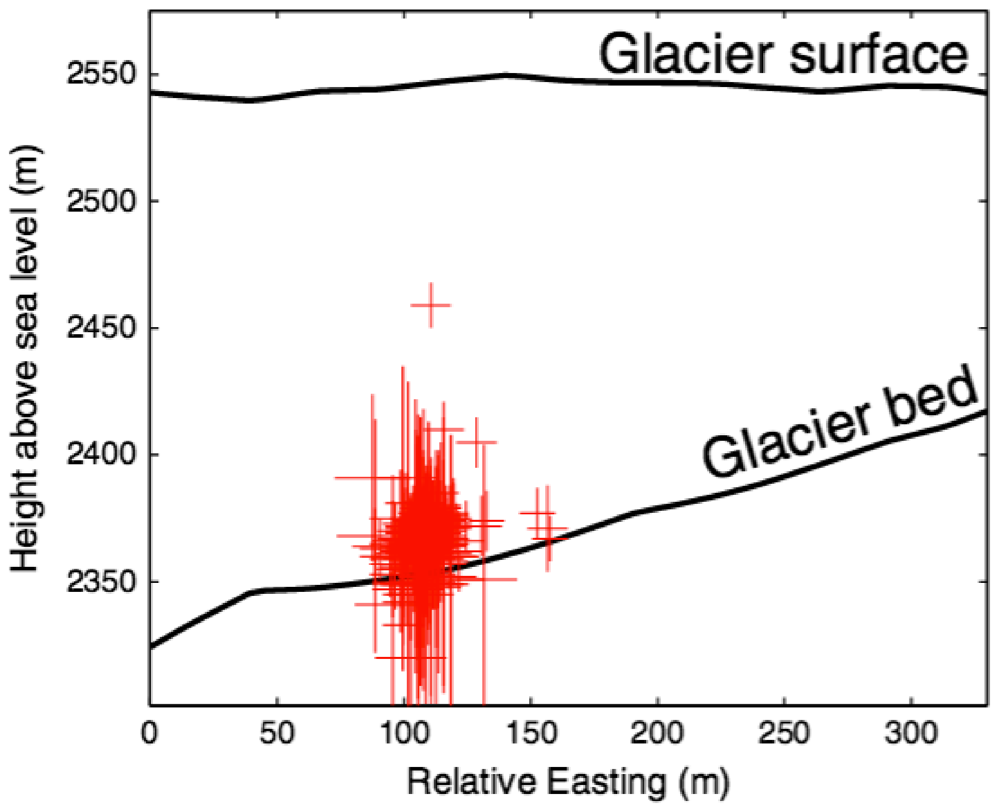
“Shallow” Normal Fault



See also
Walter et al., 2009, 2010, BSSA

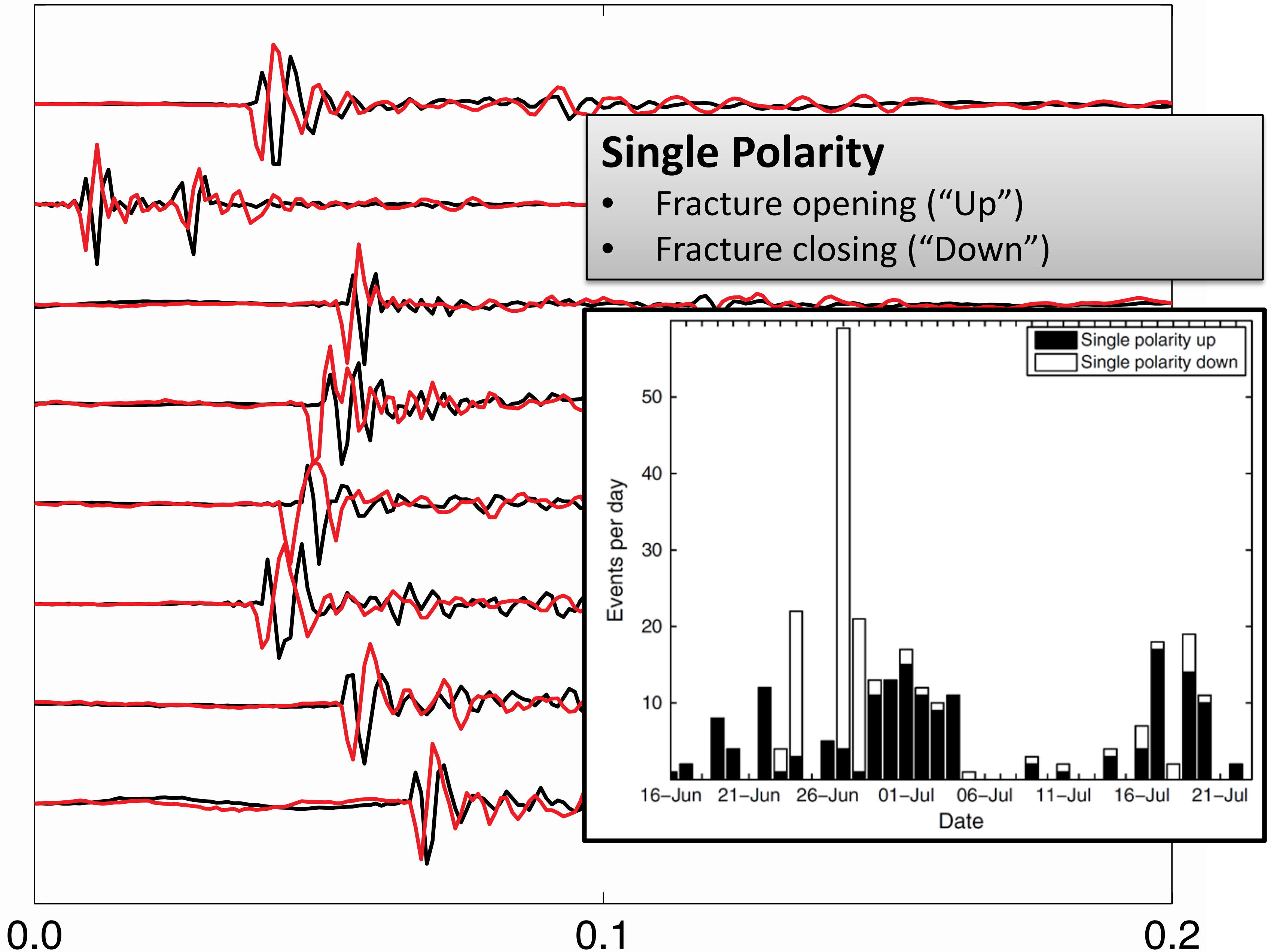
# Seismic Events Beneath Alpine Glaciers






Walter et al. (2013)



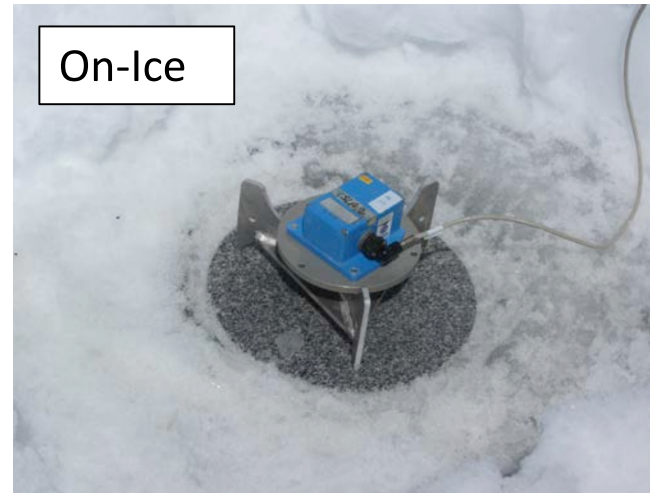

**Single Polarity**

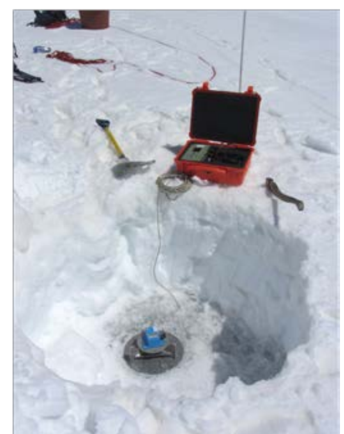
- Fracture opening (“Up”)
- Fracture closing (“Down”)

Walter et al. (2013)

# Goal: Find Stick-Slip beneath Aletschgletscher, Switzerland


500 m thick



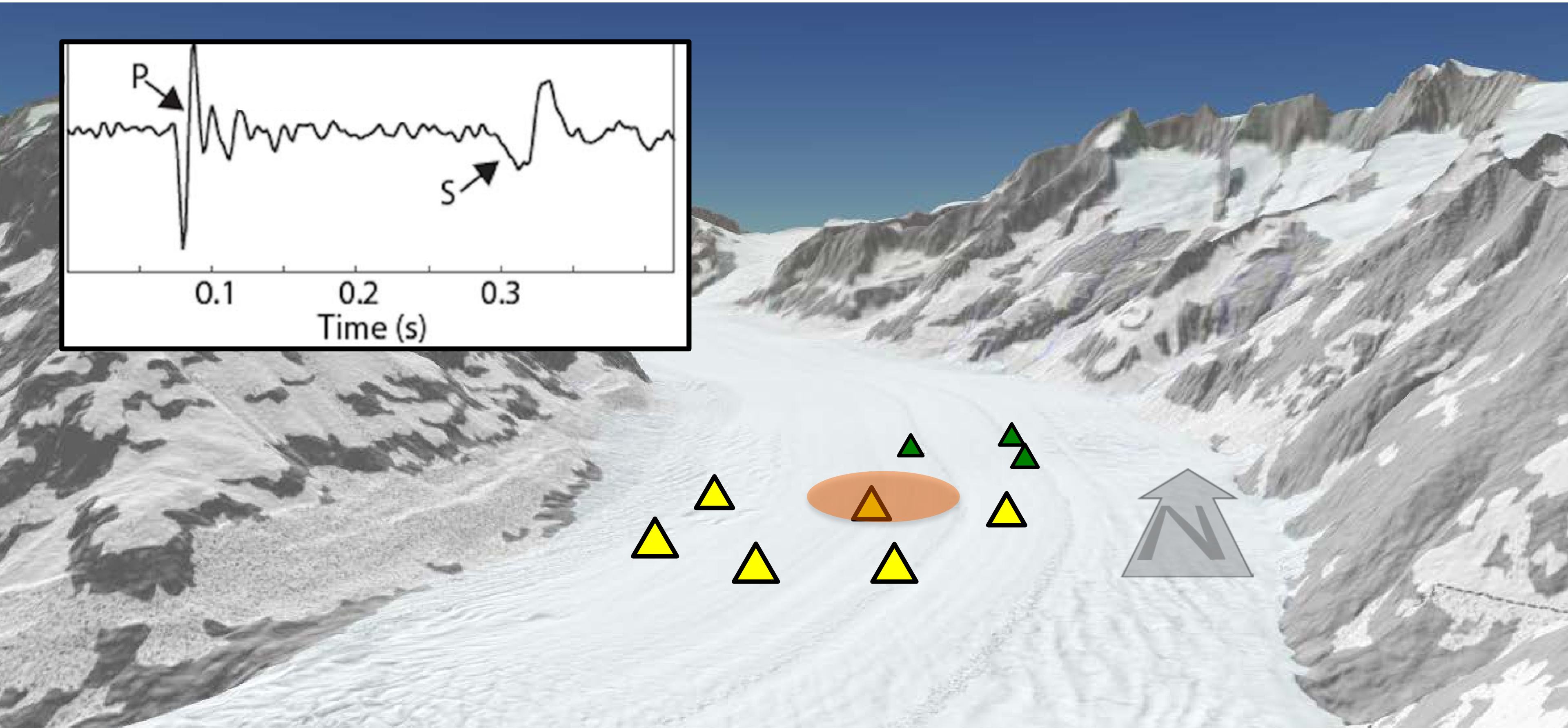



# Seismic Network on Aletschgletscher

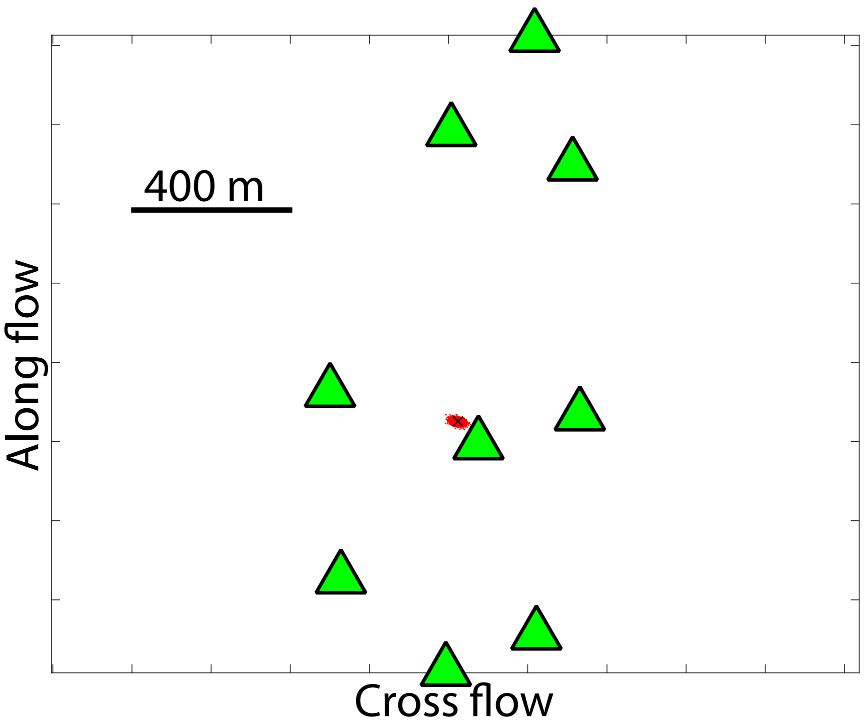
Longterm monitoring: Feb 2015 – Aug 2016

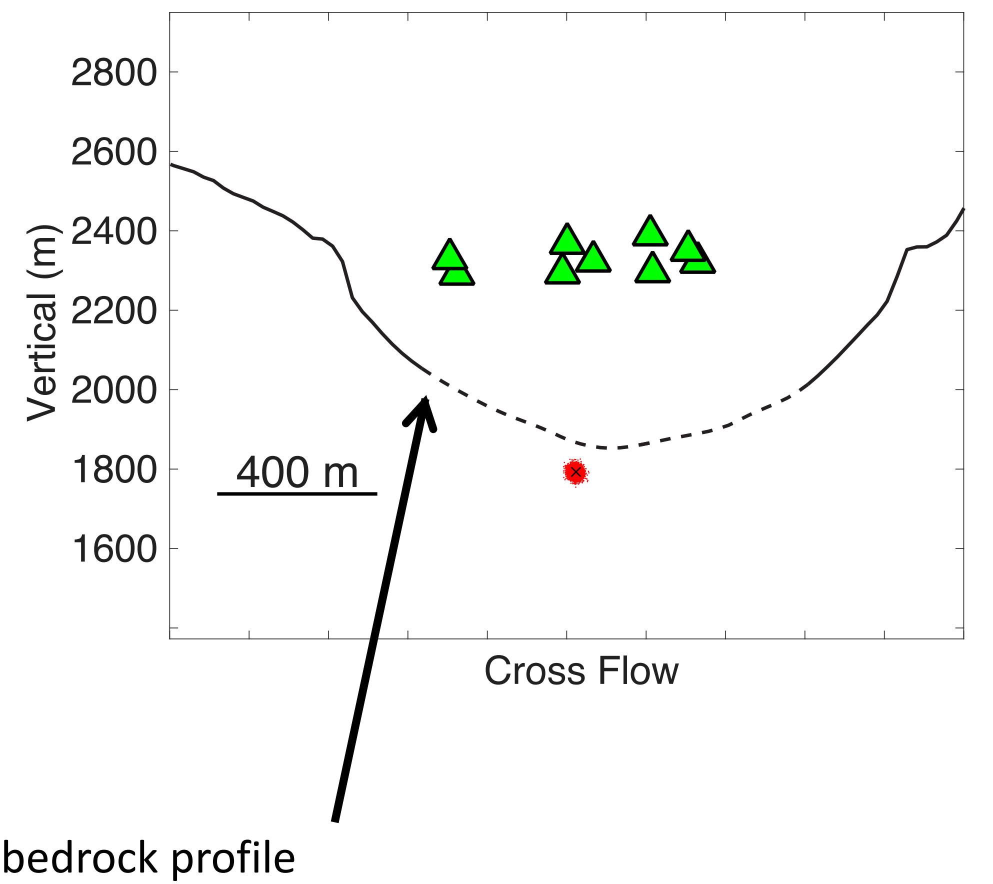
Short-term monitoring: June 2016

Area of interest:
Deep icequake cluster

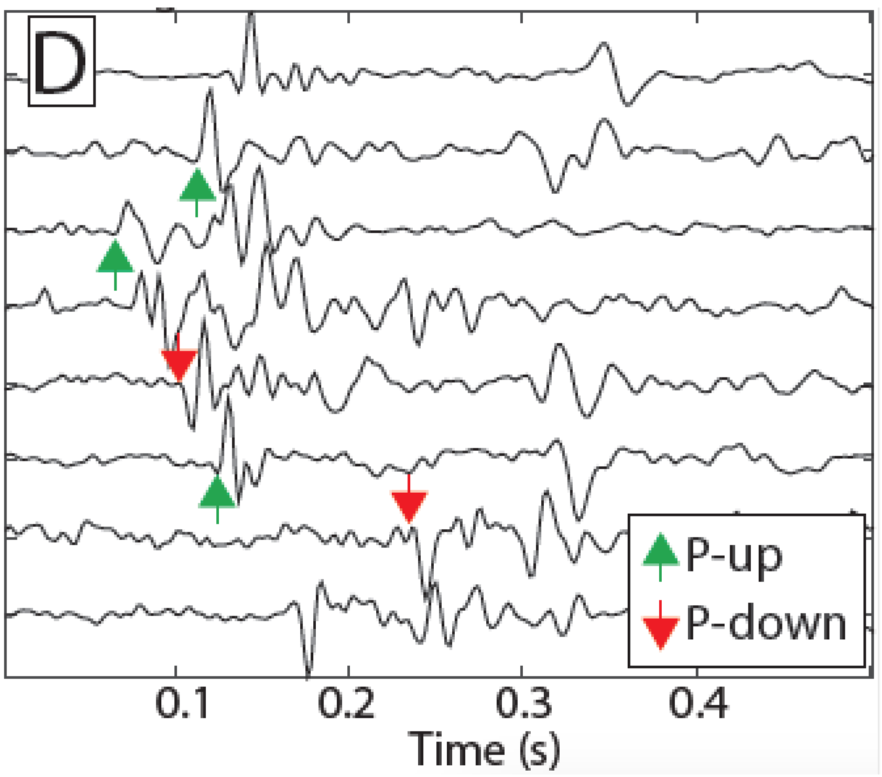


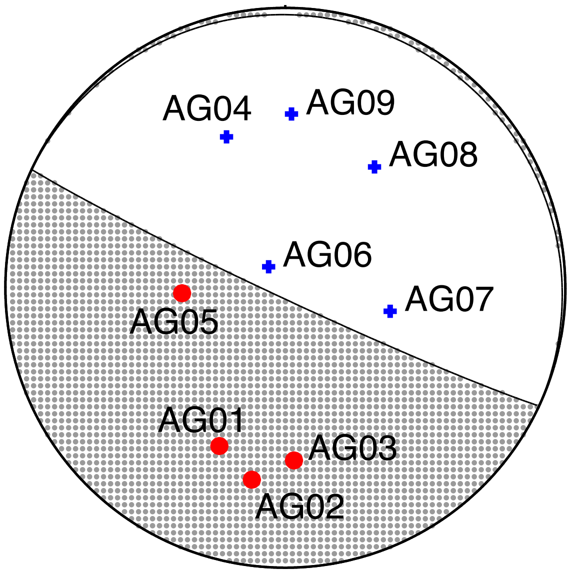
# Probabilistic Stick-Slip Location (NonLinLoc, e.g. Lomax et al., 2000)



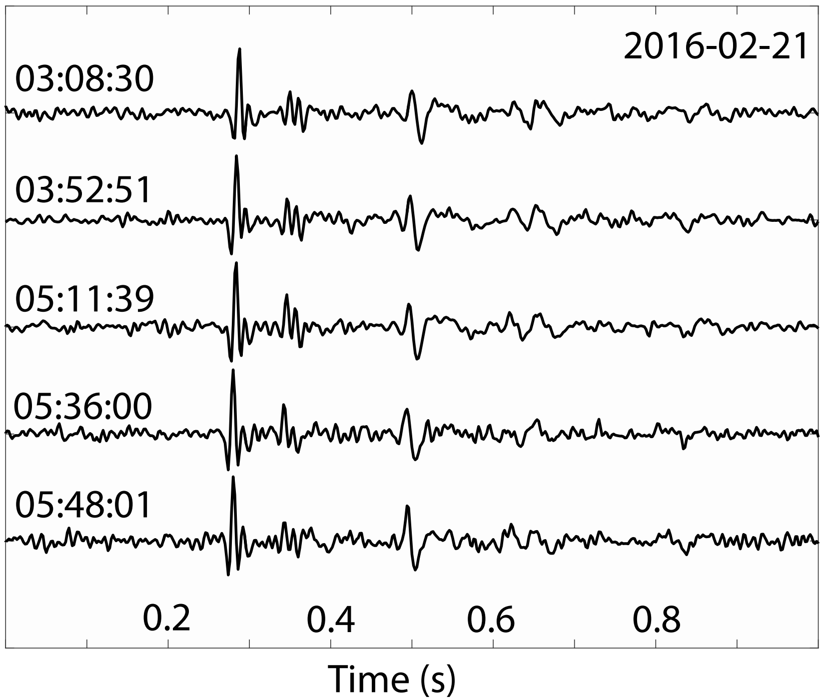



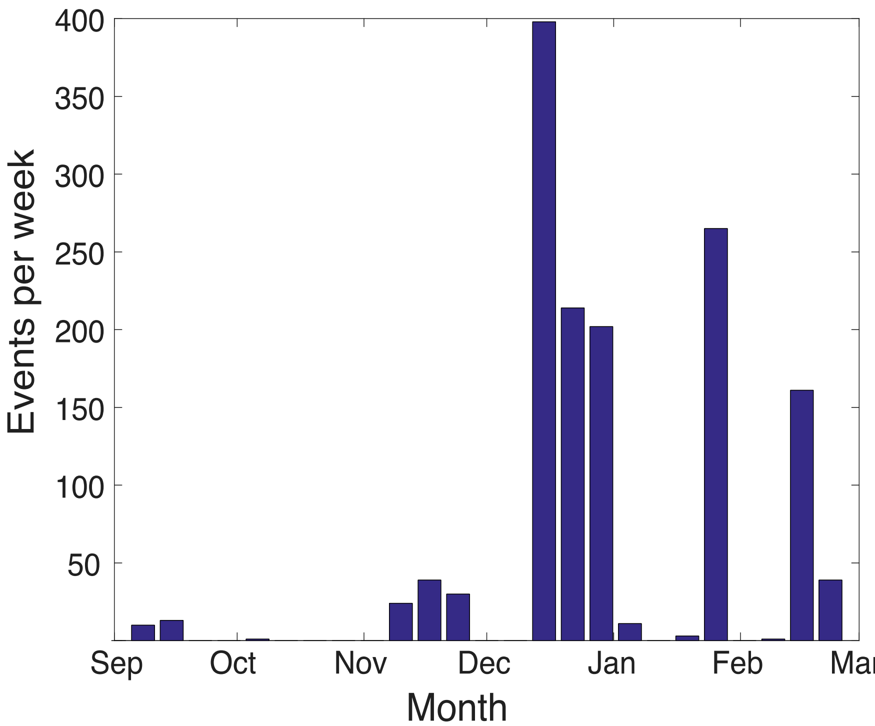
# Stick-Slip Cluster Beneath Aletschgletscher



# Stick-Slip Cluster Activity



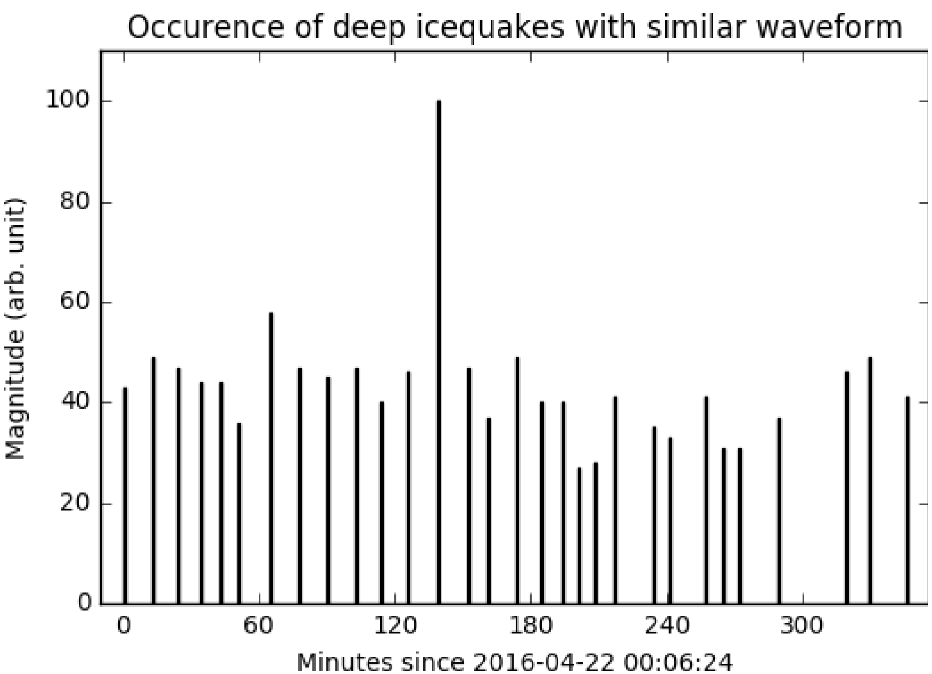



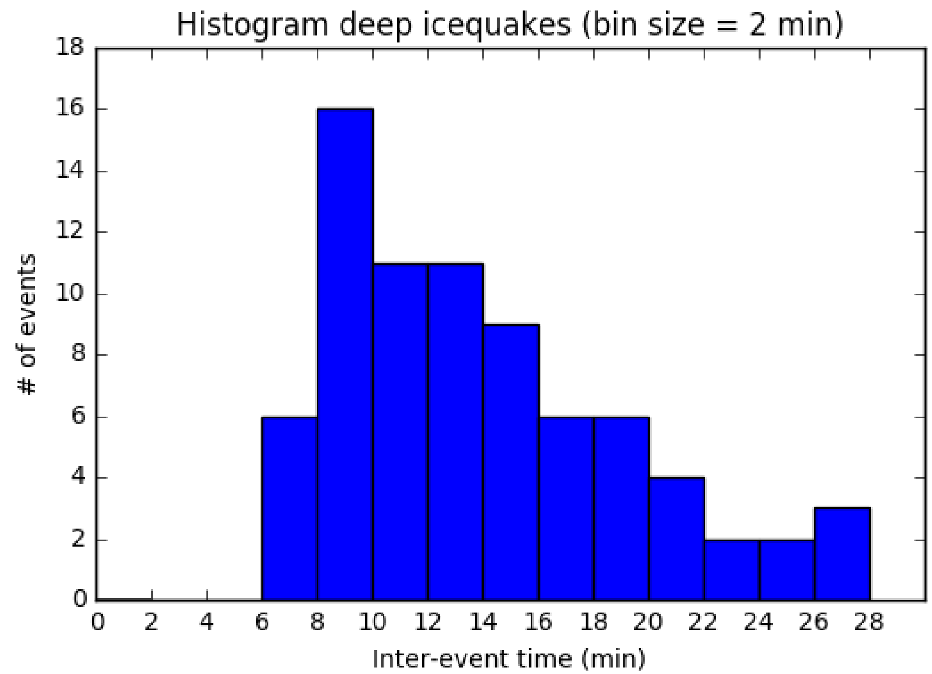

D. Gräff, PhD student at ETH Zürich

Repeating events → cross correlation search

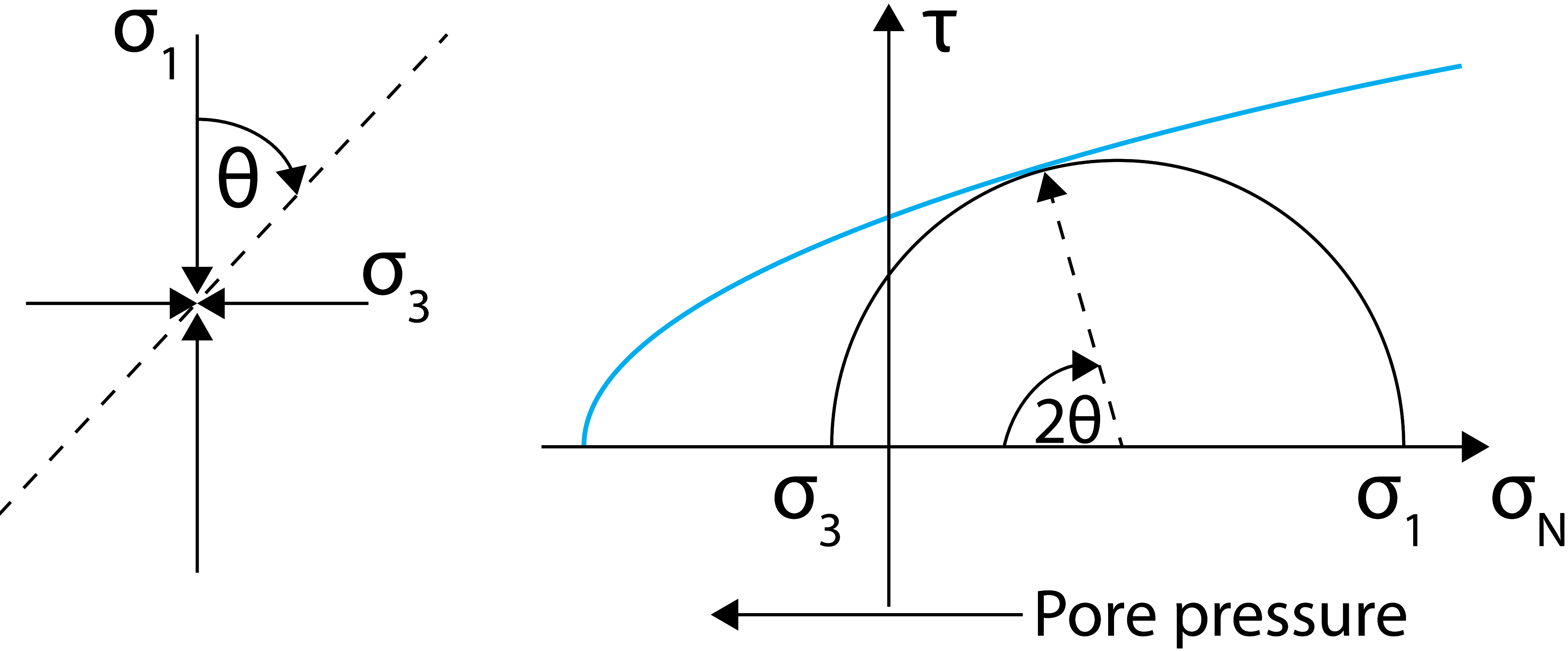
Long-term cluster activity

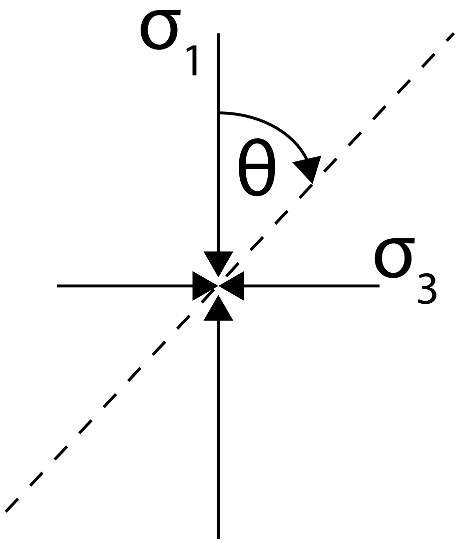
# Stick-Slip Cluster Activity



# Mohr Circle




σ1
θ
σ3

# Conclusion

Microseismic stick-slip is widespread

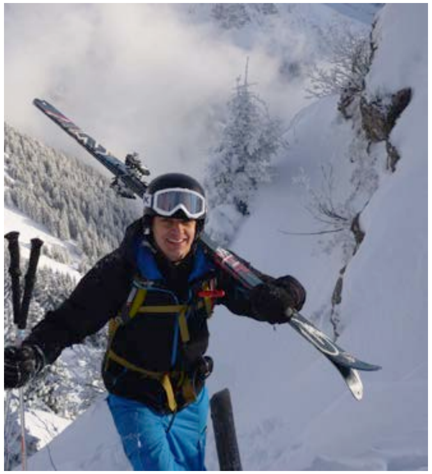
High-melt environments are difficult

Persistent asperities (order year)

Conditions for tensile / shear failure ?

# PhD Students

Fabian Lindner



Lukas Preiswerk

Dominik Gräff